

Religion and Science

Religionshistoriske Smaaskrifter I

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Within every culture we know, a critical condition sets in once the division of labour comes into force. In the material sphere it then becomes no longer possible for the individual worker to produce a work in which he can have the feeling of having worked with all his powers. What the individual can do now becomes, most often, only the production of a single part of a whole, a part that requires merely the application of a single faculty. The development of his power and his talent thereby becomes one-sided. Like his work, his very personality readily becomes but a fragment. In the spiritual sphere the division of labour means that art, science, religion, and morality, which in earlier times formed a unity, now step apart and often step into opposition to one another as contending powers. In earlier cultural periods religion stood as the great central current of life, to which all spiritual forces and drives gave their contribution, and which on the other hand yielded full satisfaction for the artistic and the intellectual need, in so far as such need stirred at all. But gradually the spiritual needs seek their satisfaction along different paths. The investigative urge is no longer satisfied by the answers that religious ideas can give to the manifold questions that life and experience may raise. Special methods are developed, special forms for posing the questions and for seeking the answers. The artistic impulse likewise goes more and more its own ways, and an independent foundation is sought for moral valuation. History shows that, in contrast to these new formations, religion has a tendency to hold fast to the notions and modes of action once established. It resists the spiritual division of labour as long as it can, and it seeks as long as possible to close its eyes to the altered conditions of spiritual life. From this follows a dividedness and a splintering in spiritual life. There is lacking that gathering and harmony, that concentration of all the forces of the mind in one common

direction, which under the earlier conditions was comparatively easy to attain. On this rests the fact that there exists a religious problem, and that such a problem will continue to exist until the spiritual life of individuals and of the race as a whole has won new gathered and gathering forms for the satisfaction of its innermost need—if indeed such new forms are possible at all.

I shall here bring forward only a few features of the intellectual side of the problem. The whole question is, for me, not exhausted by being viewed from this side. But it is of great importance to become clear about what the opposition that again and again asserts itself between religion and science really rests upon. Often one finds—and this holds of both parties that here stand over against one another—the ground of this opposition put in the wrong place. There is much superficial and dogmatic harping on science's real or supposed results on the one side, and there is on the other side much short-sightedness with respect to the change that the life of thought has in fact undergone, quite apart from the many conspicuous results. In the collisions to which the opposition leads, the onlooker does not miss honesty in the confessions made from the one side or the other. For nowadays there is, after all, comparatively little danger in making one's confession, in whatever direction it may go. By contrast, one has ample occasion to miss that intellectual integrity which does not let the mind rest before the great questions have been examined from all sides and by all the means that stand at one's disposal. This is unfortunately not a virtue that religious ethics has especially inculcated, and many of religion's opponents show, by the character of their polemic, that they too do not know the strict demands it makes. The want that an onlooker may feel here is bound up with the fact that religious debate in our days has mostly the character of agitation. On both sides it is a matter of winning adherents, of dissemination, not of deepening. New thoughts, accordingly, this modern debate has not brought either.

What I wish to bring here is likewise not new thoughts, but old thoughts that have not been allowed to come to their full due.

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If we go back to times in which religion was not merely help and consolation, but also gave answers to what thought could then ask about—times, that is, in which the whole conception of the world had a religious character—then there are two peculiarities that especially come to the fore. The whole conception of the world has an intuitive character. Existence stands as a whole whose several regions can be seen or at least represented, without its being necessary to break with what the senses immediately seemed to teach. Cosmic space, as it presented itself to sensation and imagination, contained the possibility

of localizing all the great events with which religious proclamation dealt. And when changes—especially momentous changes—entered into this world of intuitability, it was not by the arduous path of research that *the explanation* was to be sought. Such changes were explained by the intervention of personal beings in the course of events. It is intuitability, and it is the principle of personal causes (what in the widest sense may be called animism), that characterizes a specifically religious conception of the world.

But both the concept of space and the concept of cause have, under the influence of the independent urge to knowledge, undergone changes that are here of decisive importance.

1. We have a natural urge to refer everything that happens to definite places in space, and religion in its classical periods does not deny this urge. Religious representations have the character of intuition, not of concept, and their great influence, their power to grip and to fill, rests in no small part upon this. The content of the higher popular religions is a series of historical events into which the fate of men and of human life is woven—a great drama, and a drama played out upon a stage, sense-intuition, which imagination can in any case very well encompass without bursting. The whole of existence, so far as it is accessible at this stage, is taken up into this stage, and its various regions furnish places for the various scenes in the drama. The earth stands as centre, or as the lowest place, and from the visible heaven, which is the world of the deity, divine beings descend to intervene in the fate of men, perhaps themselves to suffer and die, before they ascend again into the heavenly regions, to which they thereupon open access for those who will follow them. “All men,” says Aristotle, “suppose that there are gods, and all, both Hellenes and barbarians, assign to the deity the highest place.” This utterance applies to all popular religions, however different they may otherwise be. Religious representations were woven in a natural way into men’s whole involuntarily formed conception, and there as yet existed no science that necessitated a radical change of that conception. There was no vagueness or indistinctness in the representation of the divine; this could be localized like everything else; one could point toward the place from which the divine forces welled forth, as well as to the place at which they worked. The divine thoughts were, to be sure, raised as high above man’s thoughts as heaven above earth; but heaven was precisely not raised so high above earth that intuition had to give up. Despite the difference in power and dignity, a vivid and intimate relation to the gods was nevertheless possible. No popular religion has abandoned this picture of the world. But even before science had seriously shaken the intuitability of the world, misgivings did arise among thinking minds, and the security with which men

had hitherto dwelt upon the intuitable world-picture as the frame for the whole content of religion was shaken. And this unrest was due in essential part to a deepening of the religious consciousness itself. There stirs in all religion a twofold urge: on the one hand an urge to raise the object of the religious representations as high above human and earthly conditions and forms as possible, and on the other hand precisely an urge to feel oneself one with the deity, or at least to be near it. It is an urge to stand before an Exalted One, and an urge toward an intimate relation to the Highest. Along both paths a break with the naïve intuitability of popular religion could be brought about. If the deity is to be exalted above all things, must it not then also be exalted above all spatial distinctions, above the possibility of localization and limitation? And if it is to stand in the most intimate relation to man, can it then be separated in space from the human being with whom it is to be one? A distance in space is always an external relation, however quickly it may be traversed. In the original religious documents themselves both tendencies are to be traced; there makes itself known a purely religious urge that had to lead out beyond intuitability. With this there united itself also the influence of philosophical reflection, in the case of Christianity especially the influence of Platonism. It was under the working of these various motives that Augustine could say of his God that he was “everywhere and nowhere,” and that he had his place “further within the soul than its own inmost being,” as well as “higher up than the soul knew how to climb.” But beforehand there was first a hard—both bodily and spiritual—struggle between the religious materialists (anthropomorphists), who held to absolute intuitability, and the religious spiritualists, who denied it.

In theory it was spiritualism that won. But men could not and would not wholly deny intuitability, which after all appears so clearly and unambiguously in the original religious documents and is moreover of such great pedagogical importance. In a narrative such as that of Jesus’ ascension it is indeed altogether necessary, if it is to be taken literally, to conceive heaven as a definite place, up to which the journey went. The ascension is a passage from the earthly part of space to the heavenly. The theologians of the Middle Ages then took the position of teaching that the ascension meant a passage to another kind of existence, a spiritual mode of being. To ascend to heaven means to lay aside earthly and human limitation. But this symbolizing conception did not for them exclude the literal one. Men held fast at once to intuitability and to symbolism. A passage has really taken place from physical earth to physical heaven, but this passage itself means at the same time something else. The literal and the figurative meaning of “the Higher” in contrast to “the Lower” are both held fast—though men are evidently inclined to lay

the greater weight on the figurative. It was assumed that the twofold meaning—the realistic and the idealistic, the literal and the edifying—had been laid down in the original accounts themselves. What for the original religious consciousness stood as an undivided truth, intuitible and significant at once and at a single stroke, stood for the medieval consciousness as a Both—And, at once separated and inseparable.

In Dante's great poem this standpoint prevails. The symbolic meaning is indeed here too the one that lies nearest the author's heart. But the ideal truth has a corresponding outward garb, whose reality is held fast. The unique thing in Dante's *Divina Commedia* rests, seen from this point of view, precisely on this: that the outer framework stands fast, has a massiveness and an intuitibility that the religious poetry of later ages could not attain, and that nevertheless the inwardness comes fully into its own through the symbolic interpretation that shines through. The passage from the material to the ideal is made—naïvely and grandly at once—at the very boundary of space: the outermost heaven, which encloses all things, has itself its "Where?" in the deity, whose power and love is the source of the revolution of the worlds.

But there came the time when men demanded a more prosaic answer to the question "Where?" Just as, within our experience, we determine every place by its relation to another place, so we must also assume that everywhere in the world, wherever we may think ourselves, the same holds. No absolute places or absolute boundaries can then be thought. Every intuitible object has its "Where" in the literal sense, determined by relation to other places. Therefore no world-centre and no world-boundary can be represented. No place can be found out from which the divine forces might stream forth or down. At every place the same law, or rather the same method for the determination of place, must find application. It was precisely this train of thought that led Giordano Bruno to draw his bold consequences from Copernicus' hypothesis. Already in Lucretius' famous didactic poem this train of thought was hinted at, but only far later did it become possible to hold it fast and apply it. Over against the immeasurable world of space that now opened itself, the ancient contrast between heaven and earth had to stand as vanishing. That space is in fact infinite cannot, in a certain sense, be proved. The essential thing is the new method, the new way of thinking, when the talk is of place and space. It is in a way a new organ that develops, and that thought brings with it wherever in the world of space it may betake itself. Over against this new way of thinking the old world-picture with its absolute places was untenable. The passion with which it was so long held fast, and with which men rushed upon the champions of the new world-picture, shows too that men had an inkling of what was at stake. For what was at stake was

nothing less than the surrender of the literal meaning of the religious representations. All popular religions had laid the old, absolute, and limited world-picture at their foundation. It furnished the ground and the point of support for the religious imagination and for the feeling of at-homeness with which the human spirit could regard a world in which its highest ideals could have their "natural places" just as well as the meanest material elements had theirs.

The medieval Both—And was now no longer sufficient. The events of which the biblical writings tell so reliably and intuitively must now, at decisive points, be understood figuratively. One can no longer help oneself by assuming both a real and an ideal meaning. The narrative of the ascension is here the typical example, because in it it is clearly expressed that the whole universe is affected by what happened in Palestine. A heavenly power had descended to earth and now left it again to return to its natural place.

Only very seldom does the religious consciousness seem to enter vividly into the questions that hereby arise, and that are of importance both for the historical understanding of the biblical narratives and for the characterization of the religious world-view. If the narrative of the ascension cannot be literally true, then it stands as a legend that may have its beauty and its ideal significance when we place ourselves back into the thought of the time, but that can no longer come forward with dogmatic weight. Men have thereby, on an essential point, acknowledged the legitimacy of the historical-critical method. For this method rests on the same presupposition on which Bruno's extension of the method of determining place from the earth to heaven rested, namely that in the investigation of the distant we can do nothing but apply the procedure that we have tried upon the near. We must meet the old accounts with the world-conception that we now have, even though it be ever so different from that which underlies those accounts. But it is not only for the historical method that the consequences become important. The whole significance that can be attached to religious representations must thereby become a different one. No wonder that men prefer to let the matter rest! And yet even orthodox theologians here involuntarily become symbolists and rationalists, as soon as they seriously think their way into the subject. Martensen, for example, maintains firmly that the description of the ascension is to be understood literally. But he holds that, since heaven is not any definite place, and since the meaning of the narrative can only be that Jesus now left the spatial, sensible world, one must assume that Jesus only raised himself a little way up into the air, which was to serve the apostles as a sign that he was now assuming a wholly different kind of existence from the sensible one! To this tasteless interpretation of the old account Martensen is led by his claim that the

Bible, by heaven, does not mean a place to which one comes by long roads. He starts from the symbolic meaning of the representation of heaven—and yet he will take the old narrative literally. What was a possible standpoint for a medieval theologian was an impossible standpoint in the nineteenth century, or ought to have been.

There is nothing else to do than to lay the whole weight on the symbolic meaning. It is the only way in which the old accounts can come into their full right, religiously, poetically, and historically. In every “edifying” application one in reality takes this path. Only thus can one rightly understand what has stirred in the human minds to whom these accounts are due. The “natural places” for the religious events are to be found in the interior of souls, and only there. All legends and myths are only testimonies to what has gripped minds in a certain age and set the imagination in motion. And it is the task of historical research to find out what outer events have been contributory to these spiritual occurrences, and thereby to make possible a searching psychological understanding of them. Historical research itself must again, as already remarked, start from the marks of reality that we everywhere employ where we seek to orient ourselves in the surrounding world. How far one can come along this path, and what religious significance the results of the historical investigation of the origin and development of religion may acquire, is a question by itself, into which I shall not here enter. Personally I believe that the so-called liberal theology cherishes too great expectations in this respect. — I return to the question of the influence of the thought of the immeasurability of existence upon the religious consciousness. It is by no means the case that the problem here arising should hold only for the popular religions. Every life-view, whether it be religious in the narrower sense or not, must be touched by it. Every human reflection upon the worth of life and the conditions of life must meet the consideration that human life, with its interests and ideals, its victories and its defeats, occupies only a vanishing place in the world we know. We can no longer make the whole world the stage for the drama of life. The stage upon which this drama unfolds is limited and vanishing—and what significance and worth, then, has the whole of the rest of the immeasurable universe? It is remarkable to see how little weight, as a rule, has been laid upon this question, both in philosophy and in theology, especially after the first impressions of the new conception had faded.

For Bruno it was a spiritual liberation when the narrow boundaries of the world fell before his reflection. For his poetic-philosophical conception, existence now stood as one great unity, everywhere ensouled by the same spirit and borne up by the same forces. And in a similar way Jacob Böhme too conceived the matter, once he had worked

his way out of the grievous tribulation in which he was placed when the infinity of nature had first dawned upon him in contrast to the limitation of the human world. For Holberg it still stood thus, that astronomy had now made possible a higher concept of God than the one held not only by the common man but also by many theologians. But later on the question is seldom discussed, and is preferably pushed aside. Positivists and Romantics here place themselves at bottom in much the same way as the theologians. Comte's religion of humanity introduced a symbolism in which the great universe stood as a serving background for the globe upon which human life unfolded, and even purely scientifically he declined all closer inquiry into the more distant parts of the universe. Hegel (and among us Heiberg) mocked at the "bad" infinity that extension in space presented. In Hegel's opinion men have overrated the importance of the Copernican system. Although his speculation was to give us the fundamental forms of the development of the *World-spirit*, he nevertheless maintained that the earth was the most perfect globe, because it was there that a spiritual life was found. And Grundtvig found that there was no interest in infinite space and the Copernican system, unless one started from the assumption that sun, moon, and all the stars exist for the sake of men.

There is here, on all standpoints, a contrast between world-view and life-view that may well invite reflection. Most inwardly, every serious life-view attaches itself to the narrow circle within which human tasks and ideals hold; but it is impossible that the gaze should not again and again slide out to the infinite background. We may hold fast that in the vanishing part of existence we inhabit there have developed forces and forms of life of a nobler kind than can be traced in the rest of the universe; but we do not have at our disposal forms of thought that could let these forces appear as those which determine the whole infinite course of the world. Out of that which for us has the highest value we cannot derive or understand the orbits of the globes and the surging of the world-processes. There appears to us a gulf between the realm of our ideals and the world of matter, a gulf across which no bridge can be built. One may close one's eyes to this fact, but it is more manly to look it in the face and to shape one's faith and one's hope in such a way that they can stand their test on the battlefield of reality. We must have the courage to hold fast that the stream of force and life which in its course has brought us all that we know of the great and the beautiful will also henceforth be able to make its way through the hard masses of reality, even though we can connect no meaning with the notion that these masses should exist only for its sake. —

The intuitability which is the strength of all popular religions holds not only with respect to space, but also with respect to time, and in a quite corresponding way. Yet

it is not in all popular religions that the representation of time and the representation of a historical development play an essential role. Where these representations come forward, they point to a higher ethical stage of development. It is now no longer merely the momentary and sensuous welfare of individuals or of small groups of men that is the guiding interest in the shaping of religious representations; it is the salvation of a whole people, and finally of the whole human race, that is at stake. The religious problems then concern the meaning of the whole of human history here on earth. The history of religion shows us here a peculiar contrast. For at the point in development where the urge arose to find meaning in the course of time and the march of history, there was a significant intellectual tendency that could satisfy this urge only by deadening it: the course of time had no worth at all, and it was only a matter of stopping it, of saving one's soul out of it, of passing over into a timeless being. This path development took in India, especially in Buddhism, and Greek thought, for which the unchangeable had a greater worth than the changeable, went in a kindred direction. That train of thought, on the other hand, which came to exert the greatest influence upon European humanity and upon the whole Muhammadan world, acknowledged the reality of the course of time, but found the possibility of salvation in a great course of development that leads peoples and generations to a final decision of the mighty world-struggle between good and evil powers within and outside the human world. It seems to be from the Old Persian religion that this thought especially went out into the world, whether it arose within this religion or goes further back. All eschatology, all doctrine of a development toward a last judgment and a subsequent world of glory or of perdition, seems for a very essential part to be of Persian origin. It has influenced later Judaism and through this Christianity, and it is said (according to Goldzieher) to have directly influenced Muhammadanism from its very beginning.

Just as the world-drama, as regards space, got its limited stage in the ancient world-picture, so, as regards time, it got its clear and definite place between the beginning of the world and the conclusion of the world, and the temporal distance between these two points, the beginning of the first act and the close of the last act, was just as little infinite as the spatial distance between heaven and earth. Here too popular religion has had its strength in intuitability and in the limitation that accompanies intuitability. Here man saw himself, with the whole spiritual and material weal and woe of himself and his fellows, as a link in a great course of development, determined by the common past, perhaps sighing under the torment of the present, but looking with confidence toward the redemption that the future was to bring. Popular religion has here given life a great

horizon, made possible an understanding of the individual's fate and striving from a higher point of view. And a significant advance in the historical sense was thereby made. All this, without the frame of intuitability being burst. It became possible to form clear representations of past and future, images in which all the tension and longing, as well as all the jubilation and sense of victory, that life in time can release, could find their strong, often exalted and imperishable expression.

When reflection awakens and is carried through consistently, it fares, however, with this intuitability as with the one that prevailed as regards space. The fixed boundaries, the absolute beginning and the absolute end, fall as soon as reflection applies the same method to the moments of beginning and conclusion as to the intervening moments. In place of the short time (according to Parsism 12,000 years) that the world-drama was to last, there now enters the immeasurable time, both as regards past and future. And the question then becomes how far there can be given to that span of time which man's historical consciousness can encompass such an absolute significance, such a rounded, measure-determined character, as the world-conception of popular religion so naturally presented. Can there now be any talk at all of a world-drama? The limitation in which great art shows itself becomes impossible as regards the course of the world, and therewith its dramatic character falls away.

As with the limitation of space, intuitability also presents, with the limitation of time, the possibility of a feeling of at-homeness. It goes with the world as with every single being: it has its origin, and it has its death. That the world had grown old was a common belief in the first Christian centuries; among the Jews too it was found, as is seen from the remarkable apocalyptic writing "the Fourth Book of Ezra." And one could then, amid the torments and disappointments of the time, look forward with gladness to the approaching end. The near conclusion was then no fantasy in which one now and again indulged; it stamped the whole life-view, the whole "Christianity of the New Testament."

As with space, there also takes place, as regards time, a peculiar process of inwardization in the religious consciousness itself. It seeks out beyond the distances in time, as it seeks out beyond the distances in space. Not in a distant future or past, but in present experience, will it have its object. The ideal is not to have vanished or to be merely future, but is to make itself felt in the very life that is lived in the now. Eternity is not to lie before or after time, but is to be experienced in the midst of time—as psychological experience too shows, that time disappears for us in those states that are filled with great interests and ideas. In religious mysticism, whose germs are found in the original

religious documents themselves, but which was developed further in various periods of the history of religion, the thought comes forward again and again that the true eternity can be experienced in the midst of time, when the will turns away from all that can scatter it and gathers itself about its highest object. The religious consciousness, in such forms, approaches the conception that alone becomes possible when the concept of time is thought through.

When reflection is carried through, we stand before the immeasurable time as before the immeasurable space. We must in both respects find our place, the place for our work and for the experiences we can make concerning the conditions of life. And out from this place we then pursue the course of time forward and backward, as far as we are able. But we must make ourselves familiar with the fact that there lies time and events before the point with which we can begin, and after the point with which we must end. If our interest is not limited to this span of time, which is vanishing in relation to what the strict carrying-through of the concept of time would require, then it must be a matter of holding fast the conviction that what has real worth for a surveyable time need not lose it under the continually continued course of time, even though this worth, through manifold displacements, undergoes decisive changes. Within the periods of time we can survey, we constantly see the phenomenon repeat itself, that what at first stood only as means later acquired independent worth, and that what at first stood as a final end later came to stand as means or preparation for more comprehensive values. We must give up constructing the sequence of scenes in the world-drama; we cannot decide in which act and in which scene our life falls; the very idea of a drama can be applied only to a limited time. None of the dramas that art has produced gives, moreover, by its conclusion any certainty as to how it goes with the survivors later on. And thus are we precisely placed over against the course of the world, when we let reflection prevail. Science is nothing other than steady and patient reflection.

2. With the expansion of the world-picture, as regards space and time, there followed, in that remarkable age when modern science came into being, also the demand for a stricter carrying-through of the concept of cause, and therewith a new ideal for knowledge in comparison with the one that had prevailed in antiquity and the Middle Ages. The principle was set up that, just as the events or changes whose cause one sought to find lay before one as facts in experience, so too the causes in which one found the explanation of them should be such as could be pointed out in experience. Kepler, by whom this demand was especially expressed with emphasis, had himself in his earlier writings spoken of causes that did not satisfy this demand, and which he

therefore later did not acknowledge as “*valid causes*” (*veræ causæ*). He had thus taught that the planets were held and carried in their orbits by star-spirits. But he became clear that this explanation was scientifically invalid, since these spirits could not themselves be pointed out in experience as acting in the way he had assumed. He considered, on the other hand, magnetism (which he confused with gravity) a valid cause of the planets’ being held in their orbits, because it is in fact an attracting force. However erroneous this conception was, it was nevertheless in the spirit of the new research. Later Newton showed that the force which holds the planet in its orbit follows the same law as the force which lets the flung stone fall down to the earth. It was a great and exemplary instance of a “valid cause.” Just as every place in space must be determined in relation to other places, and every moment in time in relation to other moments, so, by virtue of the principle of valid causes, every event must be determined and explained by its relation to other events that are just as demonstrable as itself. Hereby the world of experience comes to stand as a coherent whole, and nature is interpreted by nature itself, just as in a book one interprets a passage by means of other passages in the same book. The weight lies here again on the method,—on the way in which one asks, and the control one exercises upon the answer. It is a new organ that the human spirit has here developed, and like every organ it can function only in its own definite way. Thereby the tasks are set, and thereby the solutions are conditioned. And it is in connection with this a new intellectual need that has developed. Hitherto it was really only the formal sciences, logic and mathematics, that had shaped themselves into independence; now the science of experience—in its three principal forms: as natural science, psychology, and history—was in principle liberated from theological and speculative points of view. And the prospect over the new path that had been opened often awakened such enthusiasm that men proclaimed in dogmatic fashion that everything could be explained by the principle of valid causes, and that the causal series was absolutely coherent. Men took the path of construction instead of waiting until observation and experiment could show where the causes in the particular cases lay. It was a confusion of result and task, of facts and principles, of system and method. But even when men entered upon such transgressions, they were not mistaken in this, that they were on the right path. It was the spirit of research that had become conscious of its means and its powers, and therewith had also become conscious of its ideal. For it was a great ideal that now presented itself to knowledge: to shape an ever-growing number of experiences into an ever closer and more intimate unity-connection. It is an ideal that at any given time can be satisfied only approximately; it proclaims at every step an *Excelsior!*—leads ever out to greater tasks.

Dogmatic, therefore, science cannot become, when it understands itself. Already the immeasurability of space and time excludes a conclusion of our inquiry. Something new can always come forward. And that not only beyond the boundaries of the space and the time that at a given time one has been able to survey, but also within the boundaries. New differences can turn up where hitherto there seemed to be likeness or unbroken connection; an ever richer fullness of nuances and transitions can be discovered within that which seemed to be so well known. The world can constantly become greater for us, both in the great and in the small. The very concept of world comes to stand as an ideal concept that cannot be carried through, but constantly opens new horizons. And we may not even assert that existence is comprehensible or understandable, measurable for our thought. It might be compared with $\sqrt{2}$, the irrational number, which perhaps can be determined progressively through the finding of ever more decimals, but cannot receive any conclusive determination. The concept of world as a completed whole is really not at all a scientific concept. It is a concept that the religious consciousness has formed in its impatience, in order to get a sharp contrast to its representation of God, but which for science denotes nothing given, only a constant task. For science there is just as great difficulty with the concept of world as with the concept of God.

When religion in its classical periods not only gave help and consolation in the course of life, but also satisfied the urge to knowledge, it was not the principle of the “valid causes” that it called upon one to apply. Its explanation of nature was *animistic*, that is, it maintained the intervention of personal beings as ground of explanation. Just as the world as a whole was assumed to be created by an act of will, so also the several events in nature, in the life of the soul, and in history were assumed to be effected by the acting of personal beings. Especially the natural events that intervene strongly in human destinies were explained in this way. Storm and earthquake, as well as rain and fruitful seasons, found their sufficient ground in the will of Providence, which now punished or tried, now rewarded and blessed. Decisive changes in the domain of the life of the soul were explained not by psychological causes, but by divine intervention—as when the cause of Achilles’ subduing his wrath toward Agamemnon is explained by Athena’s being sent down from Olympus to talk him into reason. The march of history is interpreted by the prophets as the execution of a divine plan, which bends the hearts of princes and turns the fortunes of peoples according to its purposes. In a faded form the animistic explanation appears in the so-called teleological mode of explanation, in which one speaks in a more indefinite way of final causes, without being so consistent as to regard these ends as expressions of personal wills.

Hardly ever, however, has the animistic explanation of nature been consistently carried through. Already daily experience led to the assumption of a fixed connection between events that both lay before one as facts, so that the one came to stand as cause of the other. One let the incipient inquiry go its own ways a longer or shorter stretch, but regarded the causes it found as derived and subordinate; the real causes, the “first causes,” were personal wills. And one assumed, moreover, that even in domains where as a rule the derived causes operated, a higher cause could break through and—for the benefit of the divine purposes—bring about a change that could not be explained by the “natural” laws.

There stand here, then, two systems over against each other, two explanations of the world, the animistic and the naturalistic. There can really be no negotiation between them, for they do not acknowledge each other’s first principles. They ask in different ways and therefore also get different answers. Over against the miracle (which arises both where a “first” cause concludes the series of “derived” causes, and where it directly intervenes in the course of events) science must declare that it cannot disprove it; but it adds that, according to its principle and its method, it cannot catch sight of it at all. It can never come face to face with a miracle; for where it cannot link one event to another according to a definite law, it stands at a boundary, in any case a provisional boundary. It then does not understand the event in question; it ends in a problem. The two systems clash more and more in the human consciousness, and the victory will belong to that one of them which can unfold itself most consistently and serve as a binding tie for the greatest number of exactly determined experiences. Could existence, as it actually lies before us, be better explained in an animistic than in a naturalistic way, then the victory would be on the side of the religious world-view. But what has been attained of real and abiding understanding has not been won along that path.

It might seem as though an explanation of nature that sees in personal forces the true causes must have its great justification and necessity already in this, that man, the knowing being, is himself a personal being. Since all explanation consists in leading the hitherto unknown back to the known, one might hold that it must be the personal forces, which we immediately feel and know, that must furnish the ground for all understanding of the rest of existence. This is now, in a certain sense, also correct. We cannot leap over our own shadow, and the whole of our knowledge of the world is conditioned by the human standpoint and human nature. It is out from our own position that we orient ourselves in existence as best we can. But the decisive thing here must be which sides and properties of our nature we can make use of, when a valid and abiding understanding

is to be won. Experience teaches us, often through disappointment and pain, that it is not all our purposes and expectations that we can get satisfied. It is originally a practical interest that leads to putting definite questions to existence and demanding definite answers of it. We seek means to attain our ends, and as soon as we have discovered that we can attain our ends (those we can attain at all) only along definite paths, by the application of quite definite means, then we already stand at the causal relation that all science seeks after, the law that links one event to another. Before such a law our wishes and hopes must bow; if they do not fall away, they must often seek other outlets or more distant prospects. But through the discovery of such a law a little insight has been won into the constitution of existence, and the more such laws we can find, the better understanding we shall have. It may then gradually be a motive other than the original practical need that leads to the seeking of such laws. There can be a quite immediate satisfaction in finding connection in the hitherto scattered, whether such connection can point the way to the attainment of our ends or not. Or rather, it can itself become one of our essential ends to win understanding in the way it can actually be attained. It is precisely at this point that the significant division of labour enters, which I mentioned at the beginning of this exposition. It is, to be sure, a single side of the human personality's nature that expresses itself in the intellectual need and capacity, as it appears in its specially developed form. But what one must not overlook is that in this need and capacity the very nature of personality lays itself open to the day. A personality always presupposes an inner unity and connection between all the elements of the life of the soul, all its forces and drives. And this holds not only for the personal life considered as a whole, but also of every single endeavour to which it devotes itself with full energy. Personality cannot in any single domain deny its own peculiar essence. Within its knowledge it will therefore strive after as great and unbroken a connection as possible. It will not cease to lay stone upon stone in order to raise a building of knowledge that can stand as a counter-image of the inner connection that makes itself known—in any case as an urge—in its own essence. On this rests the deep personal satisfaction that scientific work can yield, the passion it can awaken, and the power it can have to fill out a human life. Personality and science are thus not foreign to each other, even though it is not all elements of the soul that can in equal degree co-operate in scientific work. Personality always spans more than its own intellectual work can encompass and penetrate. Besides the task which both honesty and intellectual integrity set, of procuring the most thorough orientation in the world that is possible from the position on which the individual personality stands, there is set for it moreover the task

of working together its various interests and endeavours, especially of letting the results of its intellectual work enter into as close a connection as possible with the whole life of feeling and will, with all that is object of hope and faith. That this great task is more difficult to solve than it was for other cultural periods gives no right to knock something off it from the one side or the other. We must run the intellectual line out according to its own laws, before it can in harmonious fashion be woven into the whole life of personality. It is the great task that is ever more definitely set for the people of our time—for those of them who do not dream with open eyes. The path upon which the science of experience has thus entered is therefore no accident. The empirical-scientific method has been wrung from human thought by existence itself, which will let itself be questioned, and itself answer questions, only in this way. More and more, to be sure, it becomes customary to speak of working methods and working hypotheses, and, as we shall see later, it is of great importance not to confuse the paths along which we must walk in order to catch sight of the truth with the inmost essence of existence itself. But a wholly external and indifferent relation between the path and the goal is nevertheless unthinkable. It serves to characterize existence that its laws can be found only along these definite paths. Indirectly one comes to know the nature of existence through the working hypotheses themselves.

But seen from another side too, the procedure of science has its good ground, namely if it is correct, what I sought above to show, that there is a deep analogy between this procedure and the very essence of personality. It is naturally not only this that has occasioned the method of more recent science; but it can serve to explain the deep conviction with which, again and again, despite many disappointments and mistakes, men return to this method and seek to develop it further and to apply it to new domains.

In the last generation there has happened in science various things which, to an external view, might seem to shake the reliability of the points of view that in the foregoing age were so securely relied upon. Men have accordingly been busy proclaiming a bankruptcy. They overlooked what the whole history of science shows, that in a certain sense the problems become more numerous the further our knowledge advances. Sharp problems exist at all only for those who think and inquire. And the more various experiences we make, the more tasks are set; for it is then a matter of bringing the new experiences into connection with one another and with the earlier experiences, which cannot always happen without alteration of earlier assumptions.

With the discovery of radium the principle of the conservation of energy seemed thus to be shaken. Here there showed itself a constant radiation of energy, without its

being possible to point out how this energy had been deposited in this substance. It then set new questions and gave occasion for new reflection. But it changes nothing in the old method, as regards the matter of principle. — At about the same time it was shown in biology that organisms can undergo sudden changes, whereby new types are formed. Thereby it became necessary to give up the assumption put forward by Darwin, that new types arise by small variations being favoured and developed under the influence of the struggle for existence. The mutations, the sudden changes of type, occur namely independently of the struggle for existence; only after the mutation has taken place does the question arise whether the new type can survive. But here too the presupposition is not given up, that every event must hang together with another event. The task becomes to find the inner and outer conditions under which the mutation takes place. — A third example we may take from psychology, where in the last decades attention has from various sides been drawn to sudden changes of character and personality, new formations in the psychic domain, especially within the life of feeling and will. Here too the conditions show themselves more complicated than one had perhaps for a time believed. But the points of view in principle continue to be the same; only there is greater difficulty in carrying them through.

The conditions indicated here are a wholesome warning against dogmatism in science. We have again been reminded that there is more in heaven and earth than we thought of in our philosophy. But truly scientific thinking does not need such a warning; it obeys Hamlet's counsel: to bid the new and wondrous welcome, even though our philosophy had not previously thought of it¹. — It is precisely thought and science themselves that have opened the great horizons which at the same time set us the great problems. It does not fall to those whose faith rests essentially on old traditions, which they do not always confirm by their own life-experience, to come forward with admonitions to keep the gaze open. —

A special ground for holding fast the principle of the "valid causes" lies in this, that only then does it become possible to make trial of, and get confirmation of, the cause one believes oneself to have found. When the cause of an event or a change we have experienced is sought in a preceding event that has itself appeared in experience, then there is the possibility of calling forth this preceding event again and seeing whether the presumed effect then also enters anew. If, on the other hand, one appeals to causes

¹*Hamlet* I, 5. (Horatio.) O day and night, but this is wondrous strange!
(Hamlet.) *And therefore as a stranger give it welcome.*
There are more things in heaven and earth, Horatio,
than are dreamt of in your philosophy.

that cannot themselves be experienced, directly or indirectly, it is not possible to get such a confirmation. A miracle does not let itself be repeated at will. It cannot be shown whether the personal forces or powers assumed as its cause are present anew, and present in quite the same way as the first time. Here lies the real ground of animism's intellectual barrenness. It can give expression to the wonder at the striking, the unexpected, but it cannot in the right way bid "the stranger" welcome. For to that belongs precisely this, that the new becomes, through repetition, familiar to us and is taken up into the circle of our thoughts, even though this can perhaps be done only after a change in our earlier way of thinking. Wonder is only the beginning of wisdom.

In the trial of causal explanation that is made by calling forth the presumed cause anew, the presupposition is that the very event that is to be explained has really taken place, is not a mere illusion. Wonder can often lead to a hasty assumption of reality. But here too the importance of the principle of the "valid causes" shows itself. For when we ask what mark we have of an event's reality, it appears that we are here always referred to testing whether it accords with the experiences we have previously made, fits into the whole circle of events and relations that already stand fast for us. It is the mark we quite involuntarily employ when we are in doubt as to whether something is real or not. We investigate it with our various senses, and we compare the present testimony of the senses with their earlier statements. We use the experiences of others, which we believe we can rely upon. And our doubt falls away only when this whole circle of experiences and recollections forms a coherent whole, without self-contradictions and without leaps. We apply, then, one and the same method when we test the reality of the single event, and when we seek the cause of the event thus tested. Historical criticism therefore also necessarily applies this method, whatever accounts it may stand before. It investigates how far the several testimonies agree,—whether the one is more than a mere repetition of the other,—whether what is reported accords with what experience teaches us about the course of events in the material and in the spiritual domain,—whether the content of the account cannot find its explanation in the narrator's personality and state of mind or the traditions in which he has grown up, and so on. The mark of reality is never anything single. Reality we discover by a comprehending activity of thought. A single beholding, a momentary vision, however much it may grip, contains no certainty of truth. What thus suddenly grips us can become an incitement to inquiry, to all-sided testing, and it may then show itself that it was a discovery that was made. But it contains no probative force, neither as regards the several objects each for itself, nor as regards the explanation of them. When now a critic stands before an account that contains a

miracle, the question for him will be whether it is more probable that such an event has happened, or that the narrator has been mistaken for one reason or another. Here the question will be, among other things, whether the account and its content cannot be explained out of the spiritual conditions of the time. An account is itself an event, a fact that lies before one; it is not only what is reported, but also the fact that it is reported, that can set investigation going. Historical criticism therefore applies, in its way, quite the same method that science applies in all domains.

Now there is, to be sure, a significant difference between the demonstration of the validity of general laws and the adducing of proof for the reality of a single definite historical event, namely the difference that a law must be able to demonstrate itself ever anew, whereas a historical event enters only once. With this it hangs together that it is on the whole more difficult to show the validity of the causal principle, the more special and concrete an object is. There is, with respect to the possibility of repetition, and therewith also the possibility of scientific confirmation, a whole scale of degrees from the simplest to the most complicated objects. In mechanics, physics, and chemistry it is comparatively easy to bring about repetition, to come to stand before quite the same conditions as on earlier occasions and then test whether what follows also shows itself to be the same as then. In the organic domain it is far more difficult; the state of an organism or an organ varies far more, depends on far more various conditions, than a mechanical motion or a chemical state. States of the soul vibrate and shift still more, and historical events, in which forces of nature and manifold organic and psychic conditions work together, show a minimum of repetition. Yet it is only differences of degree that in this respect make themselves felt between historical events and natural events. Not even in physical nature is there absolute repetition. Where we assume such, it rests on inexact apprehension; we do not get all the nuances and conditions with us. Therefore the method of investigating what really lies before one is in principle the same everywhere. Historical criticism has only to do with a larger and more complicated aggregate of relations and circumstances. The same holds of historical characterization, the final determination of a historical event or of a historical personality's peculiarity and significance. But this side of the matter I shall come to touch upon later on. —

We return to the miracle, and to animism as a whole, in order to call attention to the fact that within the religious consciousness itself a criticism can make itself felt of the externality and doubleness that animistic explanation, and especially the assumption of miracles, presupposes or leads to. The personal forces here appealed to are of a wholly different kind from that which they are to explain. They come from without and set

the thing in operation. There is then no inner kinship between cause and effect. The divine beings come to stand as those who operate in a world that is foreign to them. We find here the earlier-mentioned urge in the religious consciousness to raise its object as high as possible above all that is finite and earthly. But when this tendency has led to there arising an external and distant relation between the deity and the world within which man himself has his place, there is a reaction in the opposite direction. Inwardness demands its right. The deity is to be near us, to work in us “from within,” not “from without”—and then it cannot stand in a wholly external relation to “the world” and to the world-forces either. From the religious side too one cites the poet’s words: “What sort of a God would it be, who could only strike upon the world from without?” If the world were something that had independence in relation to the deity, it would itself be a god. God would have to use some of his power to work upon it. In particular, the miracle would really be a testimony to a struggle between two gods and could in any case be thought only in a polytheistic religion. For the most inward religiosity the external distinctions fall away, and all the externality that is reported of the deity’s intervention acquires in the end only figurative significance. This movement is continued in mysticism, but there is a mystical element in every inward religious consciousness. And this tendency seems, so far as one can judge, gradually to become of such preponderant importance that the miracles, which formerly were the most important means of proof for religious faith, now stand rather as hindrances—hindrances one quietly wishes away, if not on scientific, then precisely on religious grounds. In every edifying application of miracle-narratives symbolism is accordingly clearly prominent. As with space and with time, so here too, with causal explanation, there goes a tendency to symbolizing through later stages of religious consciousness. What is lacking is often precisely only this, that one becomes clearly conscious of the symbolic character.

Descartes declared in his time that in his explanation of nature he would not apply the so-called final causes, that is, explain the coming-to-be and constitution of things from the ends they are supposed to serve. It was in his eyes presumptuous, because we thereby ascribed to ourselves an acquaintance with the intention God had had in bringing forth things. It is indeed clear that such a knowledge of the ends of things is presupposed in teleological explanation. And it is presupposed still more clearly in the assumption of a miracle. The presupposition of such an assumption is, after all, that there has been sufficient occasion for the higher powers to intervene in the customary course of things. And how can that be known? Pascal held that such an occasion was at hand when a young girl was healed in the church of Port-Royal; but the opponents

of Port-Royal denied it. Only a revelation could decide the matter, and it is therefore consistent that already the medieval Church reserved to itself the right to decide whether in the particular case a miracle was present or not. In the most recent time the Catholic Church has inculcated this anew on the occasion of the many apparitions of the Madonna in the last half-century.

But it is not only the divine ends that must be known in order that one may assume a miracle. It must also be possible to decide definitely that a deviation from the law-governed order of nature has really occurred, and this really presupposes omniscience too, since it presupposes insight into the limits of nature's conformity to law, limits which science has not been able to find—and never will be able to find. —

The animistic explanation of nature, and especially the belief in miracles, is due in part to a natural urge to understand and explain, an urge that at earlier stages of development could not be satisfied by other means of thought. But the main thing was, and has more and more become, that such an explanation of nature made it easier to conceive existence and the course of things in existence as determined by that which for the human spirit stood as the highest good. Personal beings act according to ends, strive after goods for themselves and others. When it is now such beings to whose intervention the great changes in nature, in the minds of men, and in history are due, then in the end a thought of purpose stands as the decisive thing, and when man can believe that his own highest welfare is comprised in this thought, then he can feel secure amid the vicissitudes of fate. What stands for him as the highest of value is then also the last cause of what goes on in the world of reality.

But at bottom the religious view has then, by the help of animism, reinterpreted reality. The strict mark of reality, which, as shown above, consists always in the possibility of linking event to event according to the principles of the "valid causes," is disregarded, and the great problem of the relation between the ideal realm of values and real actuality is therefore not seriously posed. One does not look the matter sharply in the face, hastens with too great haste to find consolation, letting one's wishes determine what is to be truth and reality. Whatever the religion of the future may become, and whether or not the word religion will continue to fit the faith or the life-view that will develop, it will certainly in any case become its distinctive mark that it poses the great questions of life as sharply and definitely as can first become possible when the great spiritual work that science performs becomes seriously co-determining both in the questions and in the control over the answers.

In the God-concept of the higher popular religions two thoughts are fused into one:

the idea of the highest value and the idea of the absolute cause. Such a unity scientific thinking cannot reach. It finds in experience a shifting content between the valuable and the real. The Good, the True, and the Beautiful (to use this old trinity) must constantly struggle to survive and must often wrest from reality their right to be. Out from these ideas we cannot understand the whole of existence. We can perhaps find and follow a stream of light and harmony and hope that it will always be able to continue its course. But this continued course itself we cannot construct. What at a given time stands as the highest of value, the true Good, True, and Beautiful, is for a later time often only of passing significance, is a means or at least a transitional form to other values. There constantly takes place, in the great as in the small, displacements and transitions that make it impossible to say which values will continue to assert themselves. A highest value is therefore a concept that we cannot determine once and for all. The series of values we cannot think as concluded. And as it goes with this series, so it goes also with the series of causes. It constantly lengthens itself for us, and the new, future links in the series we cannot in advance derive from the preceding, just as little as we can lead the whole series back to an absolute beginning-link. As far as we see forward or backward in both series, there constantly present themselves new tasks and problems. To think both series brought to a conclusion each for itself, and moreover the one led back to the other, is to want to find the philosophers' stone. And if one begins with the belief that it has been found, then naturally all riddles become easy—all too easy—to interpret. One might ask whether scientific thinking does not then need a conclusion,—whether it cannot form an idea that denotes the last, comprehensive point of view. To this it must be answered that such a conclusion can in any case not be reached by following the series of causes forward or backward; that leads only to ever new tasks, ever new work. Not as a link in the series, whether as first or last link, as beginning or as end, can thought find a concluding idea. But the very fact that there is a causal series, a world-order that reveals itself in the conformity to law, is for thought a characteristic mark of existence, such as we know it. Therein makes itself known a great unity, a connection in the great as in the small, not only in the outer world of matter, but in our own inner being, surely further within than we ourselves can penetrate. This idea is the scientific limit-concept, which cannot receive a positive form except by the help of more or less poetic analogies. We are here cut off from employing the only procedure we have for finding positive determinations of concepts, namely comparison with other objects. That which is to be the bond between all our experiences cannot itself be bound to anything else. "That which holds the world together in its inmost core,"—to use Faust's expression,—cannot

find its expression in any of our concepts, because all our concepts are drawn from more special and limited domains. And there is added to this, as will later be more closely developed, that the significance of this limit-concept lies in the understanding that it is possible by its help to reach of the several individual existences that experience presents. The “concluding” idea contains a direction to constant work. It is set up not as an object of contemplation, but as the guiding thought whose leadership we have become conscious of. The “concluding” idea therefore really brings no conclusion, but leads us back again to our workplace in the world of experience.

Every personal life-view will now be conditioned not only by the purely intellectual motives that lead to the idea of a collection of causes, a great world-order, but especially by the place which that which stands for the personality as the highest value occupies within this world-order. It will then depend on what experiences the individual has made with respect to the struggle of values to survive. The individually different character that faith or the life-view acquires in the various personalities will depend partly on the values whose fate is object of hope and fear, partly on the experiences that have been made, or are believed to have been made, with respect to this their fate. — Upon this whole question I shall not here enter more closely. It concerns comparative religious psychology. A personal life-view must by the nature of the case have a far more special and concrete character than a scientific theory has. It arises through the combination of the scientific knowledge the individual has acquired with practical experiences of life. — Some remarks on how the relation between faith and scientific thinking shapes itself, or must shape itself, in our time shall nevertheless here be set forth.

3. Just as in nature various forces are at work, between which the natural scientist seeks to find a connection in order if possible to demonstrate their unity, so too in the human soul various, often contending forces make themselves felt, and it is a task, not only for the man of science, but for everyone, to find a connection, and if possible a harmony, among them.

Thinking and faith are two such forces or tendencies, often influencing each other, often in sharp strife, not only because some individuals will hold to thinking alone, others to faith alone, but also because they can appear as opposites in the individual's inner being and produce a cleavage there. And yet they both spring, in their inmost ground, from the same source: the human personality.

Thought seeks order and law-governed connection among the observations, and where it can find no order and law, it seeks to widen experience, to make new observations, led by the hope that what, so long as it stood isolated, seemed accidental and capricious,

may show itself to stand as a link in a whole connection, when it is held together with a greater quantity of experiences. Often it must, however, halt, because no solution can be found; the observations are either too few, or they are in their manifoldness and diversity so inexhaustible that no common point of view can be set up. Therefore thought does not on that account give up its demands and its ideals. An absolute conclusion cannot be reached; but then we end with a question, in a problem, and to that extent wonder can be not only the beginning of wisdom, but also the mood that corresponds to our wisdom's last word. Besides, there is, as we have seen, an idea that accompanies thought on its way from first to last, because it both denotes the constant task that thought sets itself, and is at the same time an expression of what the work of thought leads to when it succeeds: the idea of a great world-order that makes itself known in the laws of events and in the connection among our experiences. Thought works with a high horizon, even though this horizon at the same time always denotes the boundary of what it can reach.

While thought thus strives outward, goes from problem to problem, constantly seeks after new material and after an extension of its horizon, faith denotes a gathering, a concentration of mind and will about a single representation, a single object, which for the individual casts light over life, because he finds expressed in it what gives life value, and at the same time what secures the survival of this value through the surging of fate.

The relation between thought and faith is like the relation between a centrifugal and a centripetal force. The highest thing would be to unite the mobility of thought and its great horizon with the inwardness and concentration of faith. That is also the task that is set for every serious striving. But both in the history of the race and in the life of the individual it shows itself that this harmony is not so easy to attain. There arises again and again a strife between the opposed forces. Perhaps it is an eternal strife. As long as life lasts, there come new observations that thought is to order and comprehend, and fate brings new experiences of life to which faith is to take up a standpoint. The task of reaching a harmony between thought and faith is therefore set again and again under new conditions. The strife will then constantly be carried on under new forms.

In the nineteenth century it has thus run through various stages. At the beginning of the century Romanticism announced that the spiritual division of labour was to be over. Spiritual life was to constitute a unity, which under various forms appeared in art, in science, and in religion. The unhappy strife was to be over. What thought comprehended was to be only another expression of what art beheld and what faith surmised. In Denmark it was Henrik Steffens who proclaimed a return to a spiritual condition such as it was supposed to have been present in a bygone time that did not

know our modern oppositions and distinctions,—a time that is depicted in the following way: “In living action was impressed all that we now seek by words, all the revolutions and developments of nature, all the mysteries of religion. Life itself, fresh as it sprang from the deity, had immediately the significance that we now only surmise. Therefore all representations were living. No trace of the dividing death which in more recent times has made the sciences and poetry, and through them religion, so monstrous.” One sees that Steffens derives the difficulties in the religious domain from the emancipation and specialization of science and art. The train of thought he asserted played a great role in Danish spiritual life, from Steffens’ *Lectures* (1802) to Martensen’s *Dogmatics* (1849). But our two greatest religious personalities, Grundtvig and Kierkegaard, protested energetically against it, and their appearance has led to a sharpening of the opposition between the two spiritual forces. To this was added, in the last part of the century, the mighty upsurge of natural science, the pressing-forward of the historical method in all spiritual domains, and the reawakening critical philosophy’s testing of the validity of our representations—also of the validity of those representations in which faith procures itself expression. In the face of this state of things it might lie near to take refuge in an absolute cleavage between thought and faith, so that the task became to hold them apart from each other, since this was the only way in which both could be held fast. Under the influence of Kierkegaard’s writings and Rasmus Nielsen’s lectures I tried this way out in my youth, but both life-experience and reflection taught me that it could not lead to the goal. The two spiritual powers cannot stand wholly indifferent toward each other. Thought can make faith itself, its motives, its ideals, the representations in which it finds expression, the object of its inquiry, and on the other hand the work of thought itself springs from an urge that is akin to the one that leads to faith. There is then reason to institute a closer investigation of their mutual relation, in order thereby to recognize under what conditions the collision between them, which cannot be avoided, must be supposed likely to take place in the future. We look somewhat differently upon the whole question now than Romanticism and Positivism, the two most important tendencies of thought in the nineteenth century, did.

On two different points, both of which have been briefly touched upon in the foregoing, there shows itself an inner kinship between scientific thinking and personal life.

a. All science is a striving to find connection among scattered and heterogeneous observations. At all times men have felt an urge to this. Involuntarily, in the human spirit, the many observations are transformed into coherent images and thoughts, just

as involuntarily as the nutritive substances in the organism are transformed into flesh and blood. The difference between the various stages of development of thinking rests on how many and how exact observations one has at one's disposal, and how carefully one compares and connects them. That there must be a connection is a presupposition that involuntarily makes itself felt,—in what for us stands as childish superstition, as well as in what we call science, and which perhaps in a thousand years one will not call so. In Egyptian mythology the sun that rose in the morning was not the same as the one that had set in the evening; but they were nevertheless put in connection, the morning sun being assumed to be the son of the evening sun. The falling of the stone to the earth and the movements of the moon about the earth were found by Newton to be examples of the same law, and Ørsted found an inner connection between electricity and magnetism, which had hitherto stood as different forces. The passage-graves of the North and the domed tombs of Greek antiquity have been shown by the archaeologists in their historical connection. Psychology shows that the coming and going of our representations are subject to definite conditions, which can be formulated in general outline.

The urge to, and the joy in, finding connection is rooted, as we have already seen, deep in the very nature of personality. We call a man a personality because there makes itself known a connection among his endeavours, his moods, and his representations, and the more energetic, peculiar, and self-acquired this connection is, in the higher degree do we ascribe personality to him. Never is a personality exhausted in a single state or a single utterance, however peculiar this may be. The personal life has at once a fullness and a unity that points out beyond each single state, and the measure of personality consists partly in the compass that is spanned, partly in the inwardness with which it is comprehended. Science is now nothing other than one of the forms under which the urge to fullness and unity expresses itself, a form that in intellectual natures and scientific geniuses appears in a marked degree. It is always itself a form of personal life and is no accident in the life of the spirit, and can therefore never come to stand in a purely external relation to it. For some natures there can be bound up a great spiritual pain with having to halt at incoherent or perhaps even contradictory observations and representations, just as for every developed personality there is pain in experiences, promptings, or representations that cannot be brought into harmonious relation with the character and with the life-view as a whole. Intellectual integrity, which with all its ability and power gathers the observations that can be made and connects them in as clear a connection of thought as possible, is then also a virtue that acquires a

greater significance and necessity the more the kinship between science and personality becomes clear. Not everyone can be a man of science, but along ever more paths it becomes possible to enter into relation with the treasure of knowledge that the race has won on its way, and the individual stands answerable for the way in which such a possibility is used by him. It is a form of love of truth on which in earlier times, for good reasons, so much weight could not be laid. The love of truth could then appear only as honesty, care for agreement between confession and faith, or as personal truth, care for agreement between faith and the personality's real inner life and striving. Already the circumstance that there comes to be room for a new kind of love of truth is a testimony that the opposition between thinking and faith cannot remain standing as absolute.